

9 (long)

- (a) Explain the main differences between combinational and sequential logic. [4]
- (b) Logic values A, B, C and D represent the four bits of a binary number in the range of 0 to 15. A combinational logic circuit is to be devised which outputs 1 if the number is greater than 5 and less than 15; it is to output 0 otherwise. See Fig. 7 for details of the significance of the bits.
- (i) Construct a truth table and map the corresponding expression. [4]
- (ii) Show how to implement efficient logic systems to realise this function in terms of an expression, using minimum numbers of:
- NAND gates; [8]
- NOR gates. [8]
- (c) Explain how the Karnaugh map may be used to assist in identifying static hazards that might be present in your designs, and show how these may be corrected. [3]
- (d) It is often stated that the use of the Karnaugh map technique for simplifying logic functions leads to a solution requiring the minimum number of logic gates. Comment briefly on this proposition. [3]

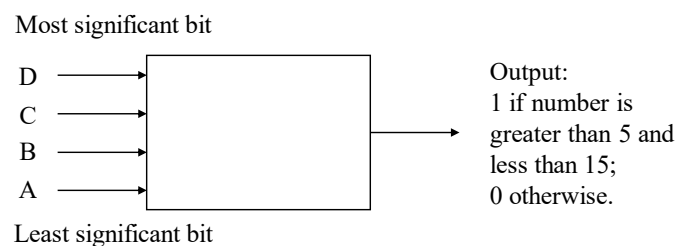


Fig. 7